

A working model of behavior leakage: a constant write, gated by context similarity

June 2026

1 Setup and claim

We study *leakage*: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appearing under other, untrained contexts. Leakage is measured as the trained–base change in the behavior’s *expression* under a bystander context, on-policy (the bystander writes its own response). The expression metric is behavior-appropriate: for a token-level behavior, the log-probability of the behavior token at the end of the bystander’s own response (the marker line, #502/#489); for open-ended behaviors, judge-scored expression rates, namely a bystander’s agreement rate with wrong factual claims minus its base-model rate for sycophancy (#509), the misaligned-and-coherent rate on open-ended probes for emergent misalignment (#521), and emission of the taught fact for fact implants (#500). Nothing below depends on which metric instantiates expression.

We represent contexts and behaviors as residual-stream activation summaries. A **context** c is summarized by $v_c \in \mathbb{R}^d$, computed from the model’s activations while it operates under c ; the construction is itself swept: candidates vary the layer, the position (last prompt token, end of the system prompt, response tokens), and the aggregation (a mean over a probe set of user questions, or the full activation cloud compared by Gaussian fits or kernel mean embeddings). A **behavior state** $v_b(c) \in \mathbb{R}^d$ summarizes the activations while the model responds under c , taken where the behavior is expressed. We also use a behavior’s *read-out direction* r_b , whose projection tracks expression (the unembedding row for the marker; a probe or contrastive direction for judge-scored behaviors, such as the difference of mean response activations under behavior-eliciting and neutral prompts), and the *write* of a training run (defined in §3). These constructions are under active comparison; the bake-offs of §7 search over them, layer included. Write f for the model’s nonlinear, unknown map from a context to the behavior state it produces there, $v_b(c) = f(c)$; nothing below assumes f is linear.

Informally, the claim is that a narrow fine-tune adds one thing to the model, a fixed *write* toward the trained behavior, and that each context receives that write in proportion to a *gate*, a base-model-computable similarity between that context and the trained one. §2 collects the literature premises; §3 and §4 state the rule and its rank-one linear special case; §5 refines the mechanism; §6 and §7 turn the rule into predictions P0–P7 and confront them with evidence; §8–§9 mark the regime boundary, the gaps, and the next experiment.

2 What the literature establishes

The premises below are each independently supported.¹

¹Two verified literature syntheses accompany this note: `docs/lora-edit-linearity-lit-review.md` (the evidence base behind L3 and §5) and `docs/icl-vs-finetuning-lit-review.md` (behind the ICL half of §5). This note keeps

L1 (weights read and write linear features).

The residual stream is a linear communication channel: every weight matrix reads a set of input directions and writes a set of output directions [1], and a rank-one matrix $uw^\top$ is the atomic read/write unit. Low-rank fine-tuning parameterizations [16] make the factorization explicit (in $\Delta W = BA$ the rows of A read and the columns of B write; a rank-1 adapter is a scalar gate on a steering vector [2]), but the factorization is a property of any weight update.

L2 (behaviors are linear write features).

Adding a single direction to the residual stream elicits a behavior (steering vectors [3, 4]); persona traits are readable as single directions at the last prompt token, and the projection predicts trait expression before generation [5]; a single “toxic persona” direction most strongly predicts and controls emergent misalignment [6]; and independently trained misaligned fine-tunes converge to one shared linear misalignment direction; EM can be induced by as few as nine rank-1 adapters [7].

L3 (narrow fine-tunes add a constant write, gated by pre-existing reads).

Narrow LoRA fine-tunes “essentially add a constant steering vector”: the added vectors have pairwise cosines near $|1|$ across prompts and tokens, including on entirely unrelated out-of-distribution text, and extracting that direction and adding it as an explicit unconditional steering vector reproduces the fine-tune’s behavior [8]. Even a trigger-gated backdoor is implemented by a constant vector added at a fixed position; the conditionality is supplied by pre-existing attention circuitry (§5). Narrow fine-tuning also leaves a context-independent activation bias on unrelated text, whose rank-1 projection carries the causal effect [9]. Conditionality, in this picture, is magnitude along a fixed write direction; no new gated circuit is built. The premise concerns narrow fine-tuning as such, not the low-rank parameterization: at matched implant strength, a LoRA and a full fine-tune of the same implant leak to bystanders indistinguishably (gap +0.00 nat, #514).

L4 (edits are rank-one key→value associations).

Linear associative memories store input–output pairs as outer products, and interference between stored pairs is governed by key overlap [10, 11]. ROME inserts a fact into a transformer by exactly the minimal rank-one update satisfying $W'k_* = v_*$, up to key-covariance whitening [12], and weight-space task vectors compose approximately additively [13].

L5 (fine-tuning is approximately kernel regression).

For small parameter updates, the function change at input x from training on x' is $\Delta f(x) \approx K(x, x') \times (\text{residual at } x')$, where K is the empirical neural tangent kernel (eNTK), $K(x, x') = \langle \nabla_\theta f(x), \nabla_\theta f(x') \rangle$ [14]; kernel dynamics approximate language-model fine-tuning in several measured regimes [15].

3 The general rule

Suppose context c' originally produces behavior state $v_{b'} = f(c')$ and we fine-tune the model so that it instead produces $v_{b''}$ under c' . Throughout, $v_{b''}$ is the *achieved* post-training state at c' : under early stopping (the band-stop recipe, which halts training inside a target source log-prob window) the achieved state undershoots the training target, so forward prediction that substitutes the target for the achieved write makes one further approximation, to be validated separately. The **write**

only the conclusions it builds on; the full claim lists and caveats live there.

premise (L2, L3) says the fine-tune contributes a near-constant write: the same output direction ($v_{b''} - v_{b'}$) everywhere, while only its magnitude varies by context. The **gate premise** (L5) says the fine-tune is a small update, small in weight and feature space rather than in behavioral effect (§8), so the per-context magnitude is first-order in the parameter change. Together, the induced change at any bystander context c is

$$\Delta v_b(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{write: } \textit{what leaks}} \cdot \underbrace{g(c)}_{\text{gate: } \textit{who leaks}}, \quad g(c) = \frac{K(c, c')}{K(c', c')}, \quad (1)$$

with $g(c') = 1$ by construction. The pre-trained map f cancels in trained–base differences, so equation (1) constrains only the edit; neither premise requires the model’s own context-to-behavior computation to be linear.

Here K is the *cross*-kernel between the evaluation read-out and the training objective, $K(c, c') = \langle \nabla_{\theta} \ell_b(c), \nabla_{\theta} \mathcal{L}_{\text{train}}(c') \rangle$, where $\ell_b(c) = \langle r_b, f(c) \rangle$ is the scalar expression score and $\mathcal{L}_{\text{train}}$ the actual training loss at the source. The cross-kernel is symmetric and positive-semidefinite only when the training gradient points along the read-out gradient, which holds approximately for a marker-only loss but not for generic next-token SFT (emergent misalignment, sycophancy); computing “the gate itself” therefore means backpropagating the training loss at c' instead of the read-out. The kernel view alone does not force a single output direction (the tangent kernel of the vector-valued f is matrix-valued); the constant-write factorization is the empirical content of L3.

The measured quantity. Measured leakage observes the write through a read-out. For a token-level behavior the read-out is close to linear: $\Delta z_{\text{marker}} = \langle W_U[\text{marker}], v_{b''} - v_{b'} \rangle g(c)$, one scalar slope per behavior times the gate, so expression behaves approximately as a per-context GLM off saturation. This statement hides approximations. The softmax correction depends on each context’s whole base distribution and can reorder contexts near saturation; that reordering is a finding in its own right, and it is why the project stores logits alongside log-probabilities. A final RMSNorm also sits between the residual stream and the unembedding, so even “linear inside” is approximate. For judge-scored behaviors the read-out is the whole generative policy, and “expression tracks $\langle r_b, \Delta v_b \rangle$ ” is an approximation licensed by L2.

Multiple pairs. Training on several (context, behavior) pairs fits a kernel system, $\Delta f(c) = \sum_i K(c, c_i) \alpha_i$, with α solving the Gram equations on the training set. Contrastive negatives are the off-diagonal entries, and their function is common-mode subtraction: persona contexts are mutually similar (pairwise cosines cluster near 0.92), so a positive-only implant writes nearly uniformly to all of them, and positive-only training indeed leaks near-uniformly (92–98% emission across bystanders), while the contrastive rig localizes (source $\sim 99.6\%$ against $\sim 11.7\%$ bystander leakage; #247, #329). The same form also qualifies our own evidence base, with a time dependence. The Gram-corrected coefficient (similarity-to-source minus weighted similarity-to-negatives) is the exact-fit limit; band-stop early stopping spectrally filters the solution, $\alpha(t) = G^{-1}(I - e^{-\eta G t}) r$, suppressing precisely the small-eigenvalue contrast components, so at our training doses the rule’s prediction sits between raw similarity and the full Gram correction. Every piece of gate evidence in §7 tests raw similarity-to-source. That it works is consistent with early stopping retaining more of the common mode than the exact-fit correction would remove, but the rule’s time-indexed predictor has never been run; both endpoints are free re-analyses on existing panels.

4 The linear rung: the rank-one special case

The gate is not directly observable without gradients, so we work down a ladder of proxies. The bottom rung linearizes everything: take the model’s context-to-behavior wiring at one layer to be

a linear map $v_b \approx Mv_c$, with the edit confined to that layer, so the trained parameters are M itself. The cross-kernel then factors as $K(c, c') = \langle r_b, g_{c'} \rangle \langle v_c, v_{c'} \rangle$, with $g_{c'}$ the output-side gradient of the training loss at the source; the scalar prefactor cancels in the gate ratio (it is positive exactly when the loss is read-out-aligned, the same caveat as the general case), and equation (1) becomes

$$\Delta v_b(c) = (v_{b''} - v_{b'}) \cdot \frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}, \quad \text{i.e.} \quad M' = M + \frac{(v_{b''} - v_{b'}) v_{c'}^\top}{\|v_{c'}\|^2}, \quad (2)$$

a rank-one read/write edit: read the trained context’s key, write the behavior delta. The same shape arrives by three independent routes: (i) the minimal-norm edit fitting the new pair, $\min \|\Delta M\|_F$ s.t. $M'v_{c'} = v_{b''}$, which is ROME’s update with whitened keys [12]; (ii) one delta-rule gradient step at the exact-fit step size, the classical outer-product update of associative memories [10, 11]; (iii) the linear-kernel case of equation (1). An adapter realizes the structure explicitly at its layer ($\Delta W h = B A h$: A reads, B writes); per L3’s #514 result the parameterization is instrumentation, not mechanism.

The reduction above requires the trained parameters to act at the layer where v_c is measured; under the other reading, M as the frozen model’s input-space Jacobian, the parameters are the whole network’s weights and the kernel is not the activation inner product, and multi-layer training violates the single-layer premise literally (§9). The Jacobian reading is useful for a different reason: it is valid only locally, so far from the trained context the curvature of f dominates and the induced shift direction itself rotates. The linear rung therefore predicts its own failure mode, direction rotation far from the source, and such a rotation implicates the local linearization before it implicates the write premise.

Each factor of equation (1) can then be relaxed independently:

(a) Gate proxies: linear $\rightarrow$ distributional $\rightarrow$ tangent kernel.

Cosine on mean activations is the linear-kernel proxy; Gaussian-KL adds second moments; MMD compares full activation clouds; the cross-kernel of §3 (one read-out backward pass per evaluation context plus one training-loss backward pass at the source) is the gate itself. The predictor bake-off has climbed this ladder implicitly, and the gains over the linear proxy have been marginal (0.79 vs 0.76 on the marker; 0.50 vs 0.49 on sycophancy), which suggests the linear proxy captures most of the available signal once the layer is right.

(b) Write relaxation: rank one $\rightarrow$ rank r .

$\Delta v_b(c) = U \text{diag}(s) V^\top v_c$: “one shared direction” weakens to “shifts live in an r -dimensional subspace,” and the SVD we already run measures the effective rank directly.

(c) Gate relaxation: dot product $\rightarrow$ learned nonlinear gate.

The published gating mechanism is pre-existing QK attention reads [8], bilinear plus softmax, so the true $g(c)$ is nonlinear in v_c . The constant write survives; predictability weakens from “computable from base cosine” to “estimable from a small sample of contexts,” the per-behavior tuning the sycophancy result already suggested.

5 Mechanism refinements: conditional behavior and in-context learning

The backdoor experiment of [8] pins down how a conditional behavior can be carried by a constant linear edit. A single steering vector, added unconditionally at a fixed position (the last token before the response), implements a Betley-style risky-backdoor task (risky advice only when a jailbreak

phrase appears in context) with validation accuracy near 1.0 on held-out in-distribution examples, even though the write knows nothing about the trigger. The proposed mechanism: value vectors on the trigger tokens already align with the risk direction in the base model, and the learned vector shifts the last token’s queries so its attention matches the keys at those trigger positions. Patching the last-token query vectors back to their base-model values kills the jailbreak, causally implicating QK attention. Scope caveat: on this task both LoRA and the steering vector fail the out-of-context-reasoning (OOCR) generalization test, so the result establishes in-distribution conditionality only.

(i) *The gate is attention-implemented.* The fine-tune’s write can be a query shift that recruits a pre-existing read circuit, with the behavior content fetched from the context’s own value vectors. The net effect at the behavior slot factorizes as write $\times$ gate only under an assumption this mechanism makes visible: the content fetched from each context’s values must itself align with one base-resident behavior direction. Where that fails, the delivered direction varies by context and the constant-write prediction (P1) fails for a mechanistic reason rather than through noise; this refinement predicts P1 violations of exactly the kind the marker arm shows (#521). When the alignment does hold, the gate is a softmax attention pattern, potentially near-binary for a discrete trigger and smoothly similarity-like only for diffuse contexts such as personas; this is the mechanistic case for ladder rung (c). (ii) *The learned write need not match the prompted behavior direction.* The learned vectors have low cosine both with the naive concept-difference vector (prompting the concept and subtracting) and with each other across seeds; many different vectors couple into the same pre-existing circuitry. The behavior’s content lives in the base model, and the edit only routes to it. This matches the marker arm’s shared direction not being the prompted-marker direction (#521), and it motivates behavior-side base-model quantities as predictors in their own right (P3).

The rule plausibly covers contexts that arrive in the prompt, not just contexts trained on. A self-attention + MLP block provably converts its context into a transient rank-1 weight patch on the MLP [17], extended to the gated, bias-free, RMSNorm blocks of the Qwen/Gemma family by [18]; in this framework’s terms, an ICL prefix performs the same write $\times$ gate operation as a fine-tune, applied for the duration of the context. It follows that the leakage rule should apply to prompt-elicited behavior shifts and that the context-side gate proxies should extend to ICL contexts. The second is already observed: base-model cosine/JS ranked marker leakage across ICL and multi-turn source contexts out of the box (#489).

The full “ICL = implicit fine-tuning” equivalence is *not* available as a premise; for pretrained LLMs it is an explicitly open hypothesis. ICL and gradient descent differ in demonstration-order sensitivity and modify output distributions differently [20]; the similarity metrics behind the original equivalence evidence [19] are passed even by untrained models; and the two differ in layer causality, since the ICL “update” to a layer depends only on lower layers while a gradient update flows through the whole model [21]. In-context behavior is therefore a proxy for fine-tuning, and the proxy carries its own validity question. What survives is a predictive link, and it is the most useful part: ICL on the same examples post-hoc recovers more than 85% of the predictions fine-tuning corrects, in-distribution [19]; a model’s own in-context inferences over a training corpus both anticipate fine-tuning’s generalization failures (the reversal curse leaves fine-tuning near zero where ICL is near ceiling on the same data) and repair them when added to the training mix [22]; and for behavioral and stylistic adaptations the entire effect of in-context demonstrations compresses to a single steering-vector-like latent direction [23]; this is the same behavior-state object that the write side of equation (1) uses. Forward prediction of fine-tuning side effects (leakage) from base-model in-context behavior is untested in the literature; that gap is P7.

6 What it implies

Each structural feature of equation (1) is a falsifiable prediction; rung-specific sharpenings are noted. Verdicts live in the table of §7.

P0 (the adapter itself decomposes into read/write).

On the linear rung the trained update is inspectable directly: per layer, the top right-singular vector of ΔW is the key the edit reads, the top left-singular vector the direction it writes [2]. For contrastive-trained adapters the kernel system predicts the key interpolates between raw $v_{c'}$ and the whitened source-minus-negatives contrast direction, moving toward the contrast direction with training dose (the two can be nearly orthogonal at panel cosines ~ 0.92 ; at band-stop doses expect an intermediate alignment); the raw- $v_{c'}$ prediction applies to positive-only artifacts (#561). Testable on stored adapters with no new training. Caveat from §5: the write may align with a circuit-coupling direction rather than the prompted behavior delta, and a mismatch is informative about mechanism; it does not by itself count against the model.

P1 (shared write direction).

The activation shift at the behavior read-out points in the same direction ($v_{b''} - v_{b'}$) for every bystander context; only the scalar gate varies. This is the write premise (L3) tested in our regime, and §5(i) names the mechanism by which it can fail. To test it, stack per-bystander shift vectors and run an SVD; the top component should dominate and match the pre-training behavior delta. Rank- r weakening: the shifts span a low-dimensional subspace.

P2 (the gate is a similarity to the trained context).

The per-bystander coefficient is $g(c) = K(c, c')/K(c', c')$. On the linear rung this is $\langle v_{c'}, v_c \rangle / \|v_{c'}\|^2$, which explains why context cosine predicts leakage and is asymmetric: the coefficient from $c' \rightarrow c$ is $\cos \theta \cdot \|v_c\| / \|v_{c'}\|$, not invariant under swapping source and bystander. The falsifiable claim is that some fixed kernel on base-model context summaries sets who leaks (positive-semidefinite only in the aligned-loss case, §3), and that the same kernel transfers across held-out panels.

P3 (the prior probability of the behavior under a context).

The context's current prior on the behavior enters through separate channels, and the design must keep them apart. (i) *Source-side residual*, formal in equation (1): the write magnitude is the gap between the trained behavior and what the source already produces, so an instructed source has residual near zero and the rule predicts little leakage even at high context similarity, where a similarity-only predictor fails. (ii) *Bystander-side read-out*, a measurement channel: the equation pushes every bystander prior-independently, but measured expression passes through the read-out anchored at the bystander's own base level, so high-prior bystanders compress $\Delta \log P$ and cross emission thresholds earlier. (iii) *Bystander-side gate*, a mechanism channel: under §5's routing picture, a bystander that already represents the behavior receives a larger fetch, and the prior becomes part of $g(c)$. Channels (ii) and (iii) separate in logit space, since compression predicts the bystander prior carries no signal for Δz once the gate is known while routing predicts it does; but the test discriminates only if the gate term is the gate itself, because with an imperfect proxy, prior-survival is equally consistent with the prior absorbing the gate's unmeasured component. A clean separation needs the cross-kernel gate.

P4 (additivity / Gram structure).

Training on two pairs solves a two-point kernel system; when the cross-kernel term is small,

leakages simply add (the activation-space analogue of task arithmetic [13]), and in general the deviation from additivity is predicted by $K(c'_1, c'_2)$.

P5 (layer locality).

The gate proxy should be taken where the behavior is computed: late layers for a next-token marker, earlier for higher-level behaviors such as sycophancy. On the general rung this restates where the tangent-kernel signal concentrates and adds no assumption. Scoring it requires an independent localization of where each behavior is computed (activation patching or layer-wise probing); scoring it by where the proxy predicts best is circular.

P6 (cross-behavior transfer).

The same update moves a different behavior at a fixed context in proportion to the projection of the write onto that behavior’s read-out direction, extending the rule from context generalization (train on (c', b) , evaluate other contexts on b) to behavior generalization (evaluate the trained context on other behaviors).

P7 (ICL proxy).

To the extent the transient-rank-1 picture of §5 holds, the base model’s in-context behavior on the training mix should forward-predict the post-training leakage map: show the training examples in-context, measure behavior expression per bystander context, and correlate with measured post-training leakage. This is the in-context analogue of the base-prior predictor (P3), and forward prediction of fine-tuning side effects from base ICL is unclaimed in the literature.

What would count as refutation. Each prediction names a fallback (rank- r write, nonlinear gate, routing), so no single miss refutes the rule; refutation requires the conjunction. Compute the cross-kernel gate exactly (no proxy), measure the write directly on a rig-clean implant (post-#561), and if write $\times$ gate still fails to rank leakage on a held-out panel, the first-order picture itself is wrong and no rung remains. Cheaper tests can disambiguate sooner: a P1 failure plus a QK-patching null (no routing at work) pins the failure on the write premise itself, and a P2 failure with the exact gate pins it on the gate premise instead of the proxy.

7 Evidence and testing program

An earlier correlational line (the cosine-predictor program, #207 and successors: behavioral DVs, mostly single-seed) tested the gate in isolation, before the framework was stated; the current program (the table) tests the structural claims directly. Where they conflict, the table is the authority.

The earlier predictor line. Base-model context similarity, the linear-rung gate of P2, rank-predicted per-bystander marker leakage at $|\rho| = 0.48\text{--}0.79$ across six experiments ($p < 10^{-7}$), holding per training step ($\rho = 0.47\text{--}0.70$); the cosine and Jensen–Shannon proxies rank-correlate at 0.94, so the bottom-rung proxies are largely interchangeable. The functional form, however, was never the proportional one: the relation is monotone but not linear, no shared slope ever fit across panels, and out-of-fold $R^2 \approx 0$. The gate orders contexts; it does not yet deliver the proportional form of equation (2). One behavioral hint sat on the write side: of firing bystander completions, 92.9% emit the marker as a tail token after an otherwise persona-faithful response (against 97.6% of source firings at the response start; #456), consistent with a fixed write whose magnitude varies by context. The table’s P1 row supersedes that reading.

Where the same predictor failed or inverted. On taught facts the geometric predictor ran with the sign opposite to the rule’s prediction ($|\rho| = 0.49$, reversed), while the base model’s prior on the fact predicted correctly (+0.27; the #500 line). This cannot be excused by the regime boundary, because taught facts are the canonical lazy rank-one case (L4): it is an in-regime P2 miss, and simultaneously as the empirical seed of P3’s channel (iii). On emergent misalignment the geometric predictor collapsed outright: the prose-only re-run found $\rho = 0.09$ (#458); the earlier $\rho = 0.75$ (#404) was a code-vs-prose plus training-recipe confound (the AestheticEM cell sat at cosine 0.9234 vs 0.9242, indistinguishable, while EM moved from 2.3% to 7.6%); and the failure carries a feature-movement signature: the assistant axis rotates 38–53°, persona discrimination collapses to $\sim 47\%$ near-uniform bystander leakage (EM-specific; benign SFT does not produce it, #125), and one EM panel reversed the correlation’s sign ($\rho = -0.59$).

The current program.

	Test	Status / result
P0	SVD of stored marker/EM adapters; compare top singular directions to the dose-interpolated contrast key and to the measured behavior delta. External replication target: the per-trigger backdoor steering vectors in [8]’s repo, trained with the same behavior under 3–4 triggers but never cross-compared	Not yet run; free on artifacts we already have.
P1	SVD of per-context activation shifts after a marker implant vs misalignment SFT (#521, controls #551)	Surprising split: misalignment SFT shifts all 14 probed contexts along one shared direction ($\cos 0.96\text{--}0.98$, $\sim$ half the spectrum; seed-stable), consistent with the convergent EM direction of [7]; the marker implant does not, and the trained context is the <i>least</i> aligned. Contrast survives controls. Benign SFT also yields one shared direction (#552, running), so the geometry may be generic to plain SFT; a positive-only marker retrain (#561, running) tests whether the contrastive rig, not the behavior, sets the geometry.
P2	Correlate base-model gate proxies with measured leakage across source $\times$ bystander panels (#502, #489; sycophancy analogue). Cross-kernel gate vs cosine: not yet run. Time-indexed contrastive predictor (§3): not yet run	Context similarity ranks marker leakage well ($ \rho \approx 0.76\text{--}0.79$ at layers 19–24) and extends to ICL/multi-turn contexts; distributional proxies give marginal gains over the linear one; on held-out persona panels geometry predicts the training-induced <i>change</i> but the structure does not transfer across panels (#553); the base-model prior beats geometric distance on instruction contexts (#532); and the fact-arm sign reversal (#500) is an in-regime miss. The magnitude half of P2 is unsettled in #521.
P3	Source side: vary how strongly the source context already elicits the behavior (instructed-marker break case). Bystander side: regress Δz on gate proxy + bystander base logit to split channel (ii) from channel (iii) (#500, #531, #532)	Growing support: the bystander’s own prior on the behavior is the most consistently supported predictor across panels; the channel-(ii)-vs-(iii) split is open. The per-slot logit storage is in place but the partial-regression test has not been run, and a clean split needs the cross-kernel gate.
P4	Joint two-source training vs sum of singletons (#520, #527, #538)	Not yet diagnostic: additivity cosine reads 0.99, but each singleton’s shift matrix is itself near rank-one, so near-parallel constants sum trivially; the confound gates fail at every tested training depth. A regime where singleton shifts vary across contexts is needed.
P5	Layer sweep of gate-proxy quality per behavior, scored against an independent localization of each behavior	Layer profiles differ by behavior as the heuristic expects (marker best at layers 19–24; sycophancy at layers 1–8, best cell MMD at layer 7), but without independent localization this is consistent, not confirmed.
P6	Behavior-generalization testbed ($B \rightarrow B'$ leakage matrix, #545); context testbed (#537); formula-as-predictor (#510)	Queued: #537 running, #545 and #510 not yet run.
P7	ICL-vs-FT equivalence persona-framed (#491): K demos in-context vs SFT on the same K, equivalence at output + representation level; plus ICL-conditioned	Queued (#491 proposed); the leakage-prediction extension to ICL source contexts already holds (#489). Literature predicts task-type dependence: $\text{ICL} > \text{FT}$ on relational classification, $\text{FT} > \text{ICL}$ on classification

The dissociation. The marker arm passes the gate half of the rule (similarity ranks its leakage) but fails the constant-write half (no shared direction, with the trained context least aligned), while the EM arm fails the gate half yet passes the constant-write half (one shared, seed-stable direction). §8 interprets this.

8 Regime of validity: lazy vs rich

Equation (1) and the entire proxy ladder are fixed-kernel, first-order pictures: they assume training adds a small edit while the model’s features stay put (the lazy regime). If training moves the kernel itself (the rich, feature-learning regime), no base-geometry predictor works even in principle, because the similarity that determines leakage is computed by features that no longer exist in the base model. This is the model’s hard scope boundary, and it is diagnosable: measure kernel or feature drift between base and trained models, or watch base-geometry predictions degrade with training dose. It also makes two questions one: “how far is leakage predictable from base geometry?” and “when does fine-tuning leave the lazy regime?”

Small in which sense? The gate premise requires the update to be small in weight and feature space (so the first-order expansion holds), not small in behavior. The two come apart because the read-out and the pre-existing circuitry amplify: a lazy-regime edit can produce a categorical behavioral change. Marker emission flips discretely when the marker logit overtakes EOS at the slot, and emergent misalignment is induced by as few as nine rank-1 adapters [7]. The conditional backdoor of §5 is a single constant vector. Behavioral effect size is therefore a poor proxy for update size. On our own dial, tripling the implant depth (band-stop window 5–12 → 14–20 nats) left the edit geometry unchanged (#538), so the narrow-fine-tune regime we operate in appears to sit inside the small-update picture well past the point where the behavior is strongly implanted. The operative boundary is feature movement, and the dose titration of §9 is the direct probe of where it sits.

One law, two regimes: a conjecture the current data complicate. Read alone, the two flagship outcomes of the earlier line are the same tool pointed at two regimes: the marker gate works, and the EM collapse (§7) carries a feature-movement signature, so marker and EM look like the lazy and rich sides of one law. The current program complicates that reading, because the two regime diagnostics dissociate (§7): under the rule, a lazy arm should pass both the gate half and the constant-write half, and a rich arm neither, yet each arm passes exactly the half the other fails. Both candidate resolutions are already being tested: the marker’s P1 failure is rig-induced (contrastive marker-only loss; #561), and the EM arm’s shared direction is the generic-SFT direction instead of a behavior write (#552); in that case write constancy is a property of plain SFT and the regime boundary sits entirely in the gate. Until those complete, “one law, two regimes” remains the best available organization of the evidence; it is not yet a settled reading.

9 Gaps and the open experiment

Several gaps are already carried by the table or earlier sections: the constant-write premise splits across arms and the rig confound is unresolved (table P1; §8); the gate has only ever been proxied (table P2; this also blocks P3’s channel split); the time-indexed contrastive predictor has not been run (§3); and only the ranking claim has evidence, never the proportional form, while the earlier line is almost entirely single-seed (§7). Gaps stated nowhere else:

- The linear rung additionally assumes $v_b \approx Mv_c$; a direct check is to regress behavior states on context summaries across many pairs before training and measure how much variance a

single linear map explains. Nobody has run this.

- The rule is stated at one layer; training updates all layers, which violates the linear rung’s single-layer premise literally (§4). P5 is a heuristic patch; no derivation exists.
- The strongest constant-steering evidence ([8, 9]) comes from one research orbit, and the published bridge from LoRA-induced trait shifts to persona directions did not survive our adversarial verification (only the prompt-elicitation projection result in [5] stands).
- Whether training strength raises the effective rank or context-dependence of the edit is unmeasured in the literature (the only published rank-raising lever is training-data diversity [9]); #538 found the geometry unchanged at a $3\times$ deeper implant.

The open experiment: dose titration across the cliff. The rule’s natural test measures the write and the gate separately, once statically at a single dose and then across a dose titration. The static half has been run (#521/#551, with #552/#561 closing controls) and produced the dissociation above, including the failure of the lazy-end rank-one prediction for the marker arm. The titration half remains open, and its aim shifts with that result: conditional on #561 resolving the rig confound, the titration tests whether the two regime diagnostics move together as dose crosses the cliff; if the marker’s P1 failure persists at low dose in a positive-only rig, the titration instead diagnoses why the lazy-end prediction fails.

Design: implant at a non-saturated anchor (source log-prob 5–10 nats below ceiling; a fully-trained anchor leaves nothing to push against, #448, and this early-stopped regime is exactly where the achieved-vs-target write approximation of §3 matters most, so the write must be measured rather than assumed). Per held-out context, on-policy and never teacher-forced (#432→#456), measure (a) the expression change (the leakage DV: write $\times$ gate through the read-out), (b) the activation shift $\Delta v_b(c)$ (the write), and (c) the base-model gate proxy. Then titrate dose along the established response (59.3% $\rightarrow$ 0.7% marker across the 10–25-step EM cliff, #139), running the SVD of the shift matrix at each dose. Prediction under the rule: near rank-one with proxy-graded magnitude at low dose, then direction rotation plus proxy collapse across the cliff. Constraints: at least 3 seeds (the earlier line is single-seed). Stretch: replace the proxy with the cross-kernel gate and measure kernel drift from base to trained as the direct lazy/rich diagnostic.

10 Assessment

The durable deliverable is the question the formula operationalizes: how far is fine-tuning generalization predictable from base-model activation geometry? The framework’s value is that it organizes the graded marker-leakage gradient (the gate at work), the uniform leakage of positive-only training (the common-mode limit of the kernel system; its corrected selectivity predictor is still untested), and, if #552 and #561 resolve the way the regime conjecture needs, the failure of the same predictor on emergent misalignment (the rich regime) under one mechanism. It also says precisely what remains: compute the cross-kernel gate rather than its proxies, run the time-indexed contrastive predictor the rule actually prescribes, locate the lazy-to-rich boundary with a dose response, and settle which arm’s P1 verdict survives the rig control. The rank-one edit is one hypothesis inside that question, the bottom rung of the ladder, and the right level of investment in the exact rank-one form is the cost of testing it, no more.

References

- [1] N. Elhage et al. A mathematical framework for transformer circuits. *Anthropic (transformer-circuits.pub)*, 2021.
- [2] J. Ward et al. Rank-1 LoRAs encode interpretable reasoning signals. arXiv:2511.06739, 2025.
- [3] A. Turner et al. Activation addition: steering language models without optimization. arXiv:2308.10248, 2023.
- [4] N. Panickssery et al. Steering Llama 2 via contrastive activation addition. arXiv:2312.06681, 2023.
- [5] R. Chen, A. Ardit, H. Sleight, O. Evans, J. Lindsey. Persona vectors: monitoring and controlling character traits in language models. arXiv:2507.21509, 2025.
- [6] M. Wang et al. Persona features control emergent misalignment. arXiv:2506.19823, 2025.
- [7] A. Soligo, E. Turner, S. Rajamanoharan, N. Nanda. Convergent linear representations of emergent misalignment. arXiv:2506.11618, 2025.
- [8] J. Engels et al. Simple mechanistic explanations for out-of-context reasoning. arXiv:2507.08218, 2025.
- [9] J. Minder et al. Narrow finetuning leaves clearly readable traces. arXiv:2510.13900, 2025.
- [10] T. Kohonen. Correlation matrix memories. *IEEE Trans. Computers*, 1972.
- [11] J. A. Anderson. A simple neural network generating an interactive memory. *Mathematical Biosciences*, 1972.
- [12] K. Meng, D. Bau, A. Andonian, Y. Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.
- [13] G. Ilharco et al. Editing models with task arithmetic. *ICLR*, 2023.
- [14] A. Jacot, F. Gabriel, C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. *NeurIPS*, 2018.
- [15] S. Malladi, A. Wettig, D. Yu, D. Chen, S. Arora. A kernel-based view of language model fine-tuning. *ICML*, 2023.
- [16] E. Hu et al. LoRA: low-rank adaptation of large language models. *ICLR*, 2022.
- [17] B. Dherin et al. Learning without training: the implicit dynamics of in-context learning. arXiv:2507.16003, 2025.
- [18] Goldwasser et al. arXiv:2511.17864, 2025 (extends [17] to gated, bias-free, RMSNorm transformer blocks).
- [19] D. Dai et al. Why can GPT learn in-context? Language models implicitly perform gradient descent as meta-optimizers. arXiv:2212.10559, 2022.
- [20] L. Shen, A. Mishra, D. Khashabi. Do pretrained transformers really learn in-context by gradient descent? arXiv:2310.08540, 2023 (ICML 2024).
- [21] G. Deutch et al. In-context learning and gradient descent revisited. arXiv:2311.07772, 2023 (NAACL 2024).
- [22] A. Lampinen et al. On the generalization of language models from in-context learning and finetuning: a controlled study. arXiv:2505.00661, 2025.
- [23] S. Liu et al. In-context vectors: making in-context learning more effective and controllable through latent space steering. arXiv:2311.06668, 2023 (ICML 2024).